PRACHI BHAVSAR

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SUMMARY

Software Engineer specializing in distributed systems, cloud-native architectures, and scalable backend services. Built and scaled systems handling **5K+ concurrent requests under sub-500ms latency**, with hands-on experience in microservices, real-time data processing, and AWS-based infrastructure. Strong foundation in algorithms, databases, object-oriented design, and operational excellence, focused on building reliable, fault-tolerant, and scalable production-grade distributed systems.

EDUCATION

Arizona State University, Tempe, Arizona — Aug 2024 - May 2026
Master of Science in Information Technology — GPA: 4.0/4.0

- Relevant Coursework: Principles of Computer & Information Technology Architecture, Information Systems Development, Advanced Database Management Systems, Analyzing Big Data, Advanced Big Data Analytics/AI, Data in the Cloud
- Graduate Course Grader: Arizona State University

Indus University, Gujarat, India — Jun 2020 - Jun 2024
Bachelor of Engineering in Information Technology — GPA: 3.9/4.0

- Relevant Coursework: Data Structures, Object-Oriented Programming, Database Management Systems, Web Technologies, Cloud Computing, Operating Systems, Software Testing & Quality Analysis, Programming for Problem Solving

TECHNICAL SKILLS

Languages: C++, Java, JavaScript, Python, SQL
Backend & Frameworks: Express.js, FastAPI, gRPC, Node.js, Spring Boot
Cloud & DevOps: AWS (EC2, Lambda, RDS, S3), CI/CD, Docker, GitHub Actions, Kubernetes, Terraform
Databases & Streaming: DynamoDB, Kafka, MongoDB, MySQL, PostgreSQL, Spark
Core Systems Concepts: Algorithms, Caching, Data Structures, Distributed Systems, Execution Planning, Object-Oriented Design, Query Optimization, Scalability
AI & Data Systems: LLMs (RAG, Transformers), PyTorch, Scikit-learn, TensorFlow, Vector Indexing
Engineering Practices: Agile, Code Reviews, Debugging, Operational Excellence, Performance Optimization, Troubleshooting, TDD

EXPERIENCE

Software Engineering Intern, **The 66th Organization**, Arizona - Remote — May 2025 - Aug 2025

- Designed and optimized scalable backend services using Java, Python, and PostgreSQL, reducing API response latency by **20% (500ms to 400ms)** through query refactoring, indexing, and execution plan optimization.
- Implemented distributed caching, selective filtering, and batching strategies with Redis, minimizing redundant evaluations under concurrent workloads.
- Engineered asynchronous event-driven data pipelines using Kafka, improving throughput and ensuring reliable large-scale data processing workflows.
- Built scalable microservices on AWS (EC2, RDS, S3) using Docker and CI/CD pipelines, achieving **99.9% production uptime**.
- Reduced debugging time by **25%** using CloudWatch and ELK monitoring tools while contributing to code reviews and operational troubleshooting.

Python Development Intern, **Sparks To Ideas,** Gujarat, India — Jun 2023 - Jul 2023

- Developed optimized Python data processing modules, improving execution efficiency by **30%** using algorithmic optimizations and memory-efficient data structures.
- Designed and implemented automated testing workflows and CI/CD pipelines, reducing deployment defects by **20%** and improving release reliability.
- Contributed to Agile development cycles through architecture discussions and peer code reviews, improving scalability and maintainability across backend systems.
- Built reusable backend components using modular design, improving extensibility and reducing long-term technical debt.

PROJECTS

LLM-Powered Q&A Search Engine

- Architected a scalable semantic search engine using FastAPI, PostgreSQL, and vector indexing, handling **5K+ concurrent retrieval queries under 500ms latency** through optimized indexing, batching, and caching strategies.
- Enhanced retrieval accuracy using embedding-based semantic search, optimized similarity scoring, and context-aware ranking pipelines.

Distributed Log Processing & Analytics Platform

- Architected a fault-tolerant distributed log processing platform using **Kafka** and microservices, reliably handling **high-volume real-time event streams** across scalable backend services.
- Engineered **cloud-native data pipelines** with **AWS S3** and **DynamoDB**, optimizing ingestion reliability, distributed storage efficiency, and **large-scale analytics processing** under concurrent workloads.

Real-Time Collaborative Code Editor

- Engineered a distributed WebSocket system implementing Operational Transformation algorithms, achieving **<100ms multi-user synchronization latency** with reliable concurrent state reconciliation.
- Ensured data consistency across concurrent users by resolving edit conflicts using Operational Transformation and real-time shared-state synchronization.

Scalable Job Board Platform

- Designed a microservices-based backend using Node.js and PostgreSQL, optimizing query execution plans and indexing strategies to support **10K+ concurrent users** on AWS infrastructure.
- Improved system scalability and reliability by implementing load-balanced services and efficient database query handling under high-traffic workloads.